

What explains variation in *Plectropomus leopardus* density

on inshore reefs of the Palm and Whitsunday regions, 2007–2018

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AIMS inshore reef monitoring, site-level extract

Summary

On inshore reefs of the Whitsunday region, sites inside no-take zones hold roughly three times the density of *P. leopardus* found on fished sites. In the Palm region, 200 km away and under the same zoning framework, no difference is detectable and the specifications disagree. That regional contrast is the central result. It is a robust association, not a measured effect of protection: every specification tried leaves the Whitsunday estimate where it is, but protection was not assigned at random and nothing here identifies a cause.

Two findings support it, three complicate it, and one earlier claim has been withdrawn.

The association is robust across every specification tried, including two that target the exposure confound. No-take sites are disproportionately sheltered, especially in Whitsunday. Adjusting for wave exposure slightly increases the estimate rather than reducing it, and restricting to sheltered sites alone gives 2.58 (1.54–4.32) against 3.21 (2.25–4.58) — the same answer with wider intervals. So measured wave exposure did not account for the association in these checks. That is weaker than ruling exposure out: EXPOSURE is a three-level categorisation that may not capture the gradient well, and the sheltered stratum is itself broad. And nothing here addresses whatever else distinguishes the sites that were zoned — reserve placement was not random.

It does not depend on any single survey. Across thirteen refits, each omitting one region-year, the Whitsunday estimate ranges 2.88 to 3.38 against a full-data value of 3.21. It also clears a multiplicity correction within its hypothesis family by an order of magnitude.

The Palm null is less settled than it looks. Under the sheltered-only restriction — the specification that handles the exposure confound by design rather than by adjustment — Palm NTR 1987 is 0.47 (0.23–0.97), below 1 though only just, where every other specification puts it between 1.1 and 1.4. Eleven sites carry it. Palm should be read as unresolved, not as null. Section 9.

No clear association was found for the prespecified habitat variables, and the environmental terms are weak or unstable. None of rugosity, live hard coral or depth — the three chosen before modelling — survives a within-between split or a multiplicity correction. Once smoothing-parameter uncertainty is admitted, turbidity, cyclone exposure and structural complexity all cease to exclude 1. The thermal term is the only environmental term whose interval still excludes 1, at about –32% across its identified range, but it rests on roughly a dozen region-year contrasts and its fitted shape is not identified at the high end. Suggestive, not established.

The negative binomial does not fully account for the zeros. Against a parametric bootstrap that refits each replicate and redraws the site effects, the data hold more zeros than the model generates ($p = 0.003$). Neither of the two sensitivities available — trimming the mostly-zero site, allowing sites their own time trajectory — moved the estimate, but no hurdle or zero-inflated model was fitted, so the misfit is recorded as an open limitation rather than a resolved one. Section 11.

How much management explains cannot be answered from this design. Protection changes within none of the 71 sites, so it and the site random intercept are estimated from the same between-site degrees of freedom and no partition can separate them. Two earlier attempts to put a number on it — 48% of variance, then 14 percentage points of deviance — have both been withdrawn as artefacts of, respectively, contrast coding and of dropping the random effect. Section 6.

1. Data and scope

637 site-year records from the AIMS inshore reef monitoring site-level extract, restricted to *P. leopardus* in the Palm and Whitsunday regions: 467 records, 71 sites, 13 region-years, 2007–2018. The panel is complete — every site was surveyed in every year its region was visited, with no attrition.

The restriction is a design choice. *P. leopardus* accounts for 73% of pooled *Plectropomus* density in Whitsunday and 39% in Palm, while the excluded Keppel and Magnetic regions are almost entirely *P. maculatus*. Including them would confound species identity with region.

Species-level data in this extract is density only. Biomass and the legal size split exist solely pooled across all three congeners, and the mixing ratio changes between years, so pooled biomass is not a usable proxy for one species. Nothing here concerns biomass or size structure.

2. Preparation

Four issues had to be handled, each a silent failure if missed. The coordinate file carries a UTF-8 byte-order mark on its header while the degree symbols in its body are not valid UTF-8, so neither a plain read nor a UTF-8 read works unmodified. The same file contains WHITUSNDAY alongside WHITSUNDAY. The protection field contains a double-space variant of NTR 1987, creating a spurious fourth level.

Most consequentially, the density values are rescaled counts and the rescaling constant is not uniform — 0.6666 in 2007 to 2017, and 0.6660 in 2018. Deriving the constant per year recovers integers exactly, with a maximum deviation of 0.00000 and no ambiguous values; assuming a single constant corrupts one year silently. Because the survey area is the same for every observation, no offset is needed in the count model: a constant offset is absorbed into the intercept.

3. Survey design

In Whitsunday, 14 of 20 fished sites are exposed while 10 of 12 long-standing reserve sites are sheltered. In Palm the three protection levels are far more comparable. The region with the large apparent protection effect is therefore the region where the naive comparison is least trustworthy — which is why the analysis below adjusts for exposure and then, separately, restricts to the sheltered stratum.

The covariates also differ sharply in the scale at which they vary. maxDHW and SSTmean carry 99% and 98% of their variance between region-years, so they are identified from roughly a dozen contrasts rather than 467 observations. Cyclone exposure carries only 18% between region-years, which puts it in a better position than the thermal terms — a statement about where its variance sits rather than a guarantee it is well estimated. This governs what the model can honestly claim about each.

4. What the raw data show

The two regions behave differently and consistently. In Whitsunday the two reserve categories sit well above fished sites in every survey year. In Palm the raw summaries show no clear separation between the three protection levels in any year, and all three decline over the series. These are

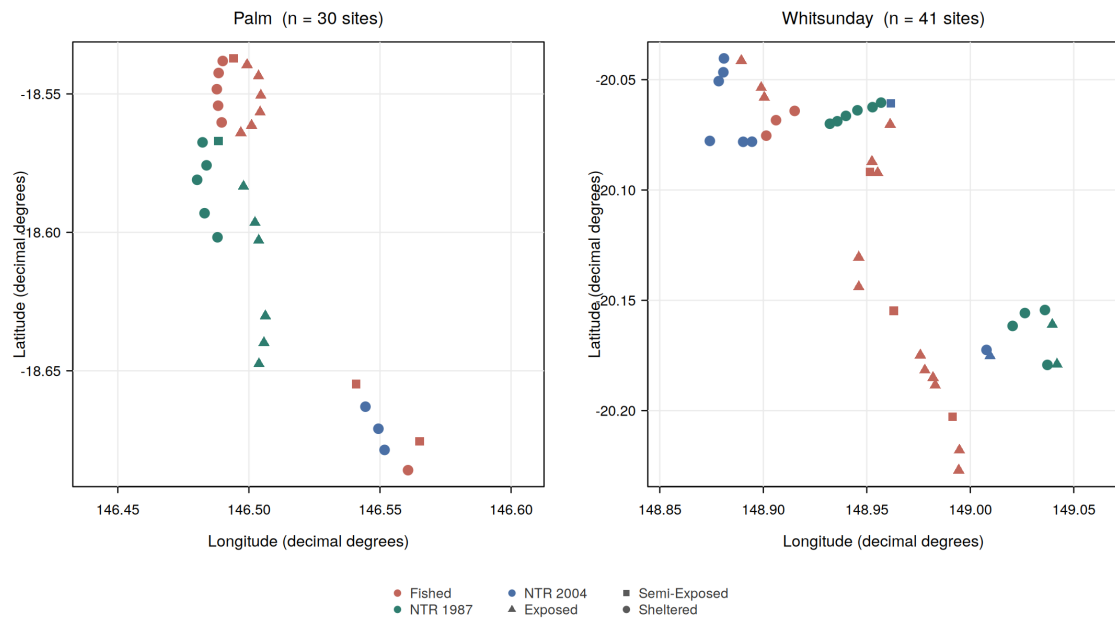


Figure 1. Sites by protection status and wave exposure. Reserves were not placed at random with respect to exposure, and the imbalance is far worse in Whitsunday than in Palm.

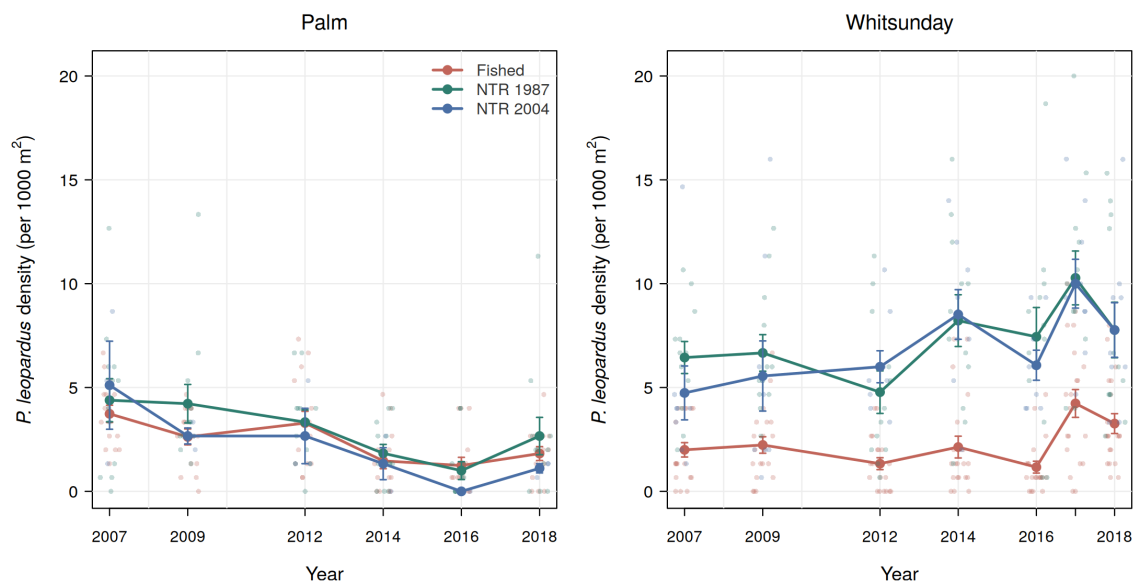


Figure 2. Density by year and protection status. Faint points are sites; heavy lines are means with standard errors.

unadjusted means: whether a difference exists in Palm is a question for the model, and the answer there is unresolved rather than null.

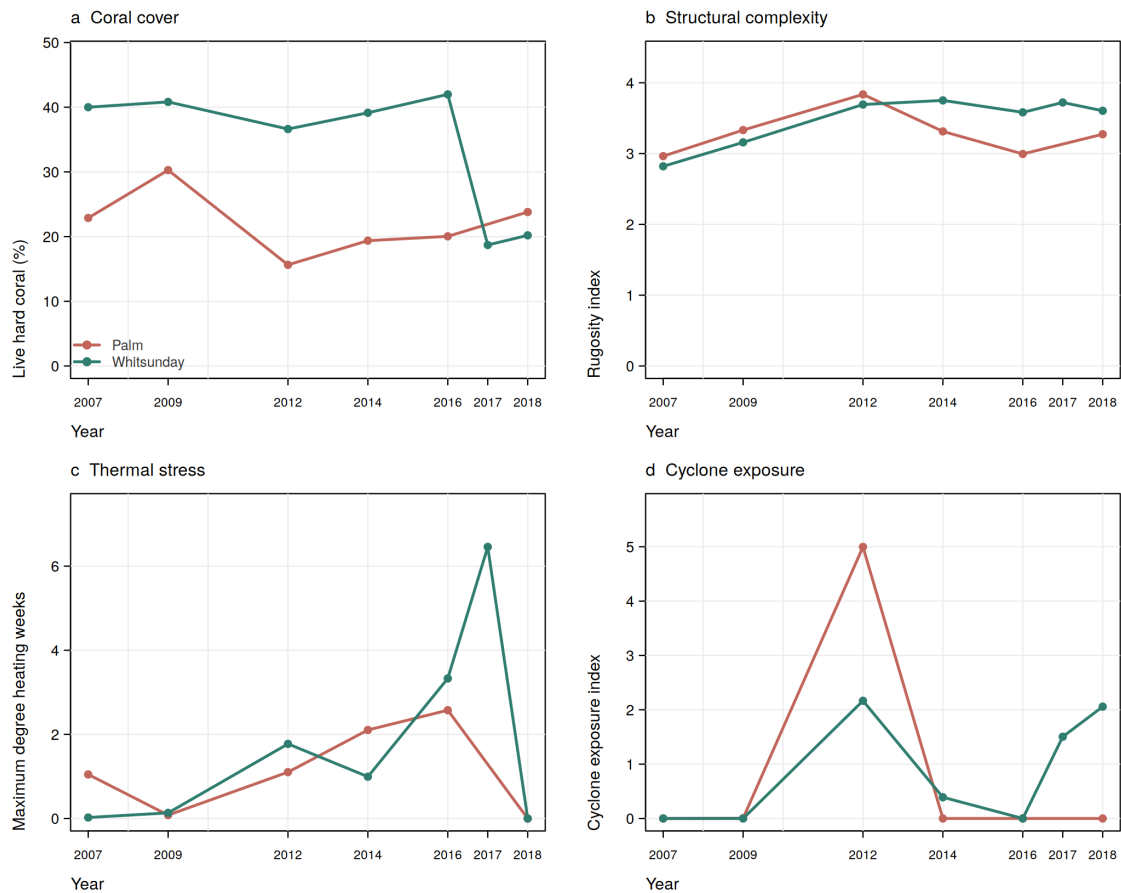


Figure 3. Habitat condition and disturbance exposure, region means. Palm was not surveyed in 2017.

The 2017 Whitsunday survey stands out. Live hard coral fell from 42% to 19%, thermal stress reached 6.5 degree heating weeks, and cyclone exposure was elevated — yet mean rugosity did not fall (3.58 to 3.72). Coral cover and structural complexity moved independently, which is what happens when thermal mortality kills coral tissue but leaves the skeleton standing. Any analysis treating coral cover as a proxy for habitat structure would misread that year.

5. The model

A negative binomial mixed model with a site-level random intercept, fitted by REML in *mgcv* using a random-effect basis. Variance-to-mean ratio in the raw counts is 5.11, so a Poisson is not tenable; the estimated dispersion parameter is 3.78 in the baseline model.

Model adequacy was checked with randomised quantile residuals, which are exactly standard normal under a correct model and therefore interpretable on a QQ plot in a way deviance residuals are not for small counts. The baseline residuals had mean -0.008 and standard deviation 0.969.

The baseline diagnostics showed clear structure in the region-year residuals, so year effects were made region-specific in the second model. That reduced the range of region-year mean residuals from $-0.65 \dots +0.50$ to $-0.17 \dots +0.16$, and raised deviance explained from 49.4% to 60.2%.

Two model diagnostics that were not run in the first pass are worth stating. Concurvity — the GAM analogue of collinearity — is high for the terms that vary at region-year level: worst-case 0.951 for the thermal smooth and 0.926 for turbidity, against the region-specific year trends. Their

coefficients are therefore one possible split of shared variation rather than independently identified effects. And a basis-dimension check flags the year smooths and turbidity. For the year smooths this is not fixable by raising the basis, since Palm has only six distinct survey years and genuine year-to-year jumps are not smooth in any basis; Model B, which saturates time, is the specification that answers it. For turbidity the flag is a real limitation.

Depth is fitted as a linear term rather than a smooth. It is constant within every site in this subset, so a penalised smooth of it was fully aliased with the site random effect — concurvity exactly 1.000. As a linear term it gives identical estimates with no aliasing, because the smooth had already collapsed to a straight line. Omitting depth altogether raises the Whitsunday estimate to 3.62, so it is doing real adjustment work and is kept.

One conflict, resolved by fitting two models

Region-year fixed effects give the best possible adjustment for time, but they absorb 99% of maxDHW's variance and make a thermal effect inestimable. The two requirements cannot both be met.

Model A uses region-specific smooth year trends, leaving year-to-year deviation available to the environmental covariates. Model B saturates time with region-year as a factor and drops the thermal terms as unidentifiable.

Environmental effects are read from A only, protection from both. The two agree on protection to within 0.02 on every contrast — close, but not to two decimal places, as an earlier version claimed.

6. How much does management explain? (Q2)

Not as a unique variance share. This design does not yield one, and the rest of this section is about why that is the answer rather than a failure to find one.

Two earlier attempts have been withdrawn. The first decomposed the variance of the linear predictor into per-block shares and reported management at 0.481; refitting the identical model under sum-to-zero rather than treatment contrasts moves that to 0.391 and space/time from 0.121 to 0.345, with identical fitted values and log-likelihood, so the number was describing the parameterisation. The second measured management by removing it from a model with no site random effect and reported 14.09 percentage points of deviance explained; with no random effect present the management terms absorb every between-site difference that happens to align with zoning, so that figure measures confounding as much as management. Both are set out in the decisions log.

The obstacle is structural. Protection changes within none of the 71 sites. Its entire contribution is between-site, and a site random intercept is also entirely between-site; the two draw on the same degrees of freedom. Remove management and the random effect absorbs it; remove the random effect and management absorbs everything else fixed about a site. The consequence is specific: there is no unique, assumption-invariant variance share to report. Any number that appears to separate them rests on an assumption the data do not test — a choice of contrast coding, or the removal of the random effect — and a different, equally defensible assumption gives a different number. That is a statement about what this design identifies, not a claim that the question is unanswerable in principle: protection varying within sites, or external information about how sites were selected for zoning, would change it.

What is reported instead is two things, neither of which is a share.

Drop-one-block at fixed dispersion. Each block is removed and the fall in deviance explained recorded, with intervals from a site-level cluster bootstrap. One correction from the previous version: `nb()` re-estimates the dispersion parameter for every model, so a reduced model can absorb its lost structure into a smaller theta and the two deviances are then measured on different scales. Every model here is fitted at the full model's theta. The results are descriptive and non-additive — the drops

do not sum to anything and are not unique variance shares — and the management row sits near zero for the structural reason above, which is not evidence of no effect.

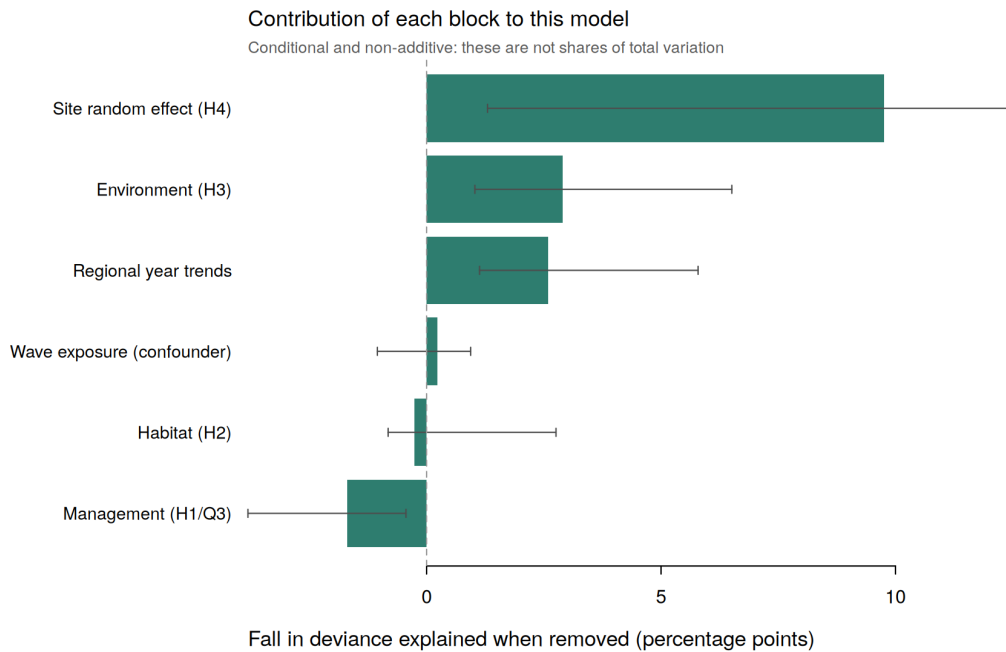


Figure 4. Fall in deviance explained when each block is removed, at fixed dispersion, with 95% intervals from 200 site-level bootstrap resamples. Conditional and non-additive: not shares of total variation.

Exploratory out-of-sample prediction to unseen sites. What is scored is the marginal predictive density of individual site-year counts at sites the model has not seen — marginal in the sense that the unknown site effect is integrated out rather than estimated. Ten folds split by site, so a site is never in both training and test. Removing management costs 48.4 in log predictive density (site-clustered SE 11.7, $z = 4.14$), a margin the other blocks do not reach in this comparison. Whether they contribute little or ten folds over 71 sites simply lack the resolution to detect it is not something this comparison settles, so the reading is that only management shows a clear signal here — not that the others have been shown to contribute nothing. Wave exposure makes prediction slightly worse, which is what an adjustment term carried for confounding control rather than for prediction can do.

The z is descriptive, not a formal test. It is the ratio of a cross-validated difference to a clustered standard error, computed on folds fixed by one seed, across five blocks compared without adjustment for that comparison. It indicates how large the difference is relative to its site-to-site variability. It is not a p -value in disguise and no threshold is applied to it.

Two corrections were applied to an earlier version of this comparison, and both mattered. The random effect of an unseen site is now integrated over rather than set to zero: for a site the model has never seen the intercept is unknown, so the predictive density is the integral of the negative binomial density against $N(0, \sigma^2)$, not its value at zero — under a log link the density at the median of that distribution is not its mean. The integral is taken by 20-node Gauss-Hermite quadrature with log-sum-exp averaging, and σ is re-estimated inside each training fold from that fold's own fit, so nothing leaks from the held-out sites; across the ten folds it ranged 0.271 to 0.339. And the standard error is now clustered on the 71 sites — the difference summed within each site before the spread is taken across site totals — rather than treating all 467 held-out observations as independent.

The integration was much the larger of the two. Management's loss in log predictive density fell from 87.3 to 48.4 and the total held-out log predictive density rose from -1232.3 to -1214.7 ;

clustering the standard error moved the z from 4.79 to 4.14. Two blocks changed sign under the correction, environment and the regional year trends, but neither is distinguishable from zero either way.

This remains exploratory, observational and not causal, whatever its z . It is a predictive statement about sites the model has not seen. It does not partition variance, the entries do not sum to the model's performance, and it does not license a claim that management contributes more variance than site identity, since that would require a unique, assumption-invariant share this design does not yield. A zoning label may predict an unseen site well because of what protection does, or because of whatever the zoning process selected for, and this analysis cannot tell those apart. The site random effect is deliberately not among the blocks compared: it is integrated out of every prediction, so its row would measure nothing and would invite exactly the comparison the design cannot support.

H4 expected unexplained site-level variation to be large relative to the measured covariates. The site random effect has the largest point estimate of any block, at 9.76 percentage points — and that is the correct way to state it. Its bootstrap interval, 1.30 to 12.43, overlaps those of environment (1.03 to 6.51) and the regional year trends (1.13 to 5.79), so the ordering among the blocks is not established; only the point estimate is highest. H4 is supported in direction rather than demonstrated as a ranking.

7. Habitat and environment

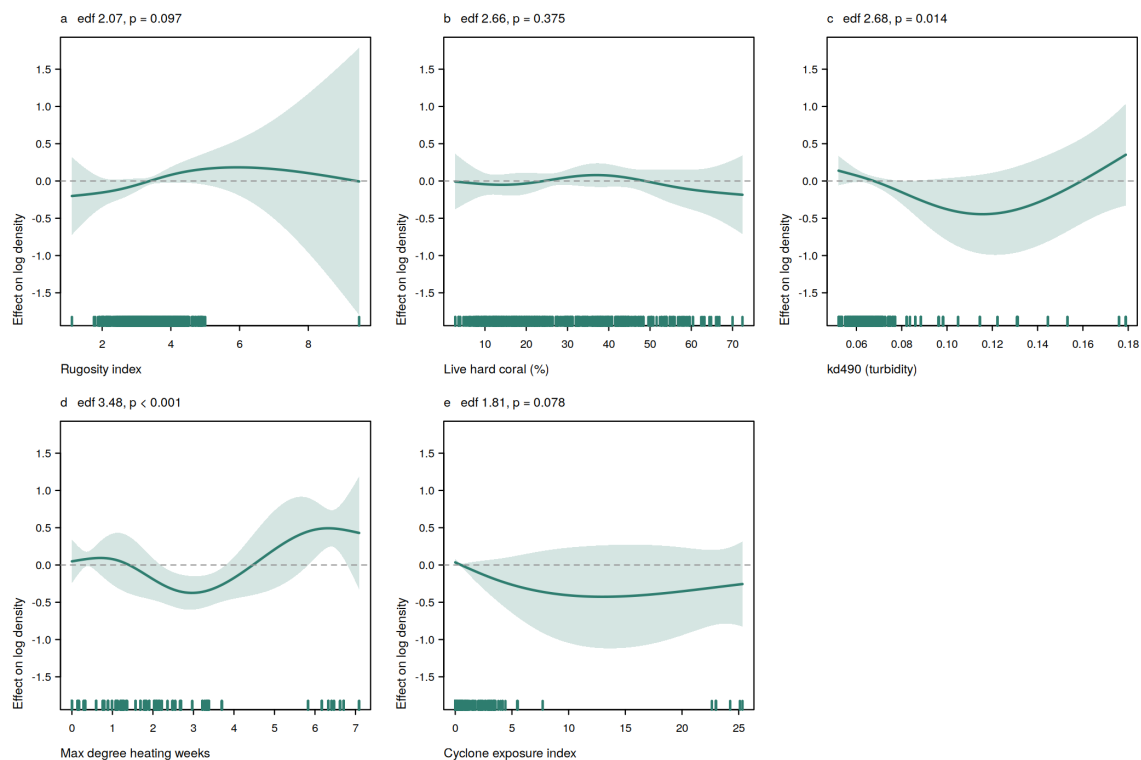


Figure 5. Partial effects from Model A, with effective degrees of freedom and approximate p-values.

None of the three habitat terms is significant: rugosity $p = 0.097$, live hard coral $p = 0.375$, depth $p = 0.099$, and none survives a Holm correction within the habitat family. Once site identity and region-specific time are accounted for, no clear association is found for any of the three variables chosen before modelling. That is a null for these variables as this extract measures them — three site-level summaries, two of them habitat condition and one bathymetry — and not a finding that habitat does not predict density.

That null needed checking twice, and the second check changed it. Expressed as a magnitude rather than a p-value, moving rugosity from its 10th to its 90th percentile gives a ratio of 1.27, and on the conditional covariance matrix its interval was 1.04–1.54, which excludes 1 — an apparently firmer result than $p = 0.097$ suggests. Two things dissolve it. A pooled habitat coefficient conflates two different claims: that sites with more structure hold more fish, and that a site which gains structure gains fish. Splitting rugosity into a site mean and a within-site deviation separates them, and neither component excludes 1: between sites 1.28 (0.88–1.85), within site 1.13 (0.98–1.31). And the pooled interval itself widens to 0.98–1.64 once the uncertainty in the smoothing parameters is admitted, so it no longer excludes 1 either. The estimate looked precise for two independent reasons, neither of them a signal.

Q4 asks whether the habitat association runs with structural complexity, with live coral cover, or with neither. On the within-site estimates — the ones immune to confounding by any fixed characteristic of a site — the answer is neither.

The thermal term was the strongest, but its fitted curve rises above roughly 4 degree heating weeks, which would mean heat stress increases coral trout density. The rug shows a gap between 4 and 5.8, and Whitsunday 2017 is the only region-year above 4 — 41 of 467 observations.

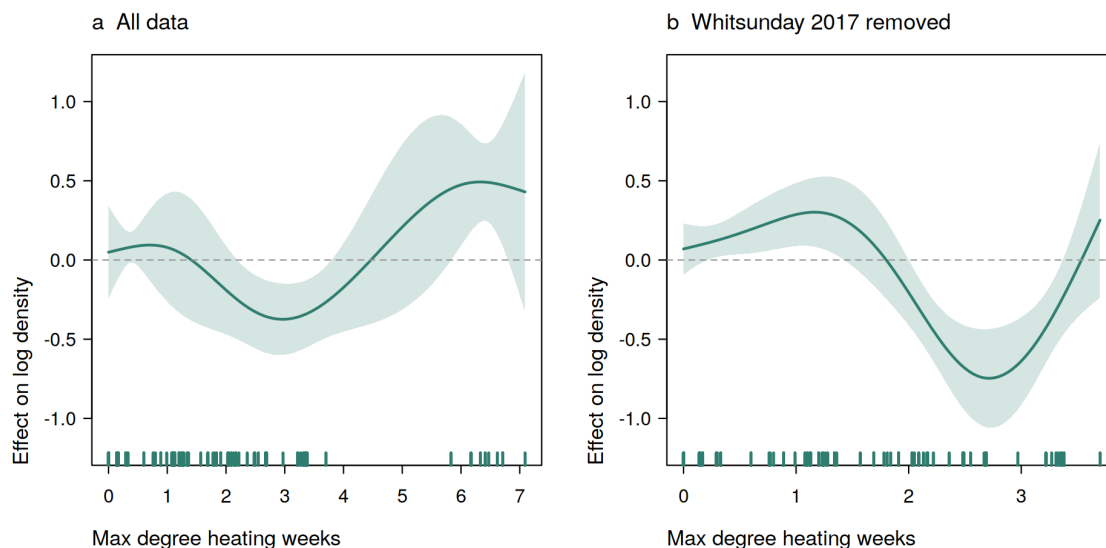


Figure 6. The thermal smooth with and without the Whitsunday 2017 survey.

The dip at moderate thermal stress survives removal of that survey and is if anything deeper. The rising limb does not, and is a boundary artefact of the penalised spline in both fits. So the shape at moderate thermal stress is at least not an artefact of one survey — which supports reading the association as suggestive, alongside the identification limits in the next paragraph. Nothing about the high end is identified and it should not be quoted at all.

8. How large is each association?

Significance and magnitude are different questions. Each covariate below is moved from its 10th to its 90th percentile with everything else held fixed, which is what the primary research question actually asks. Because the contrast is a difference of two rows of the same design matrix, every term not involving that covariate cancels exactly, including the site random effect.

All intervals here use the unconditional covariance matrix. The default treats the smoothing parameters as though they were known rather than estimated, which makes every interval on a smooth too narrow. Admitting that uncertainty changes the conclusions: rugosity, turbidity and cyclone

exposure all stop excluding 1. For the protection contrasts, which are parametric, it widens the Whitsunday interval from 2.28–4.52 to 2.25–4.58 and changes nothing.

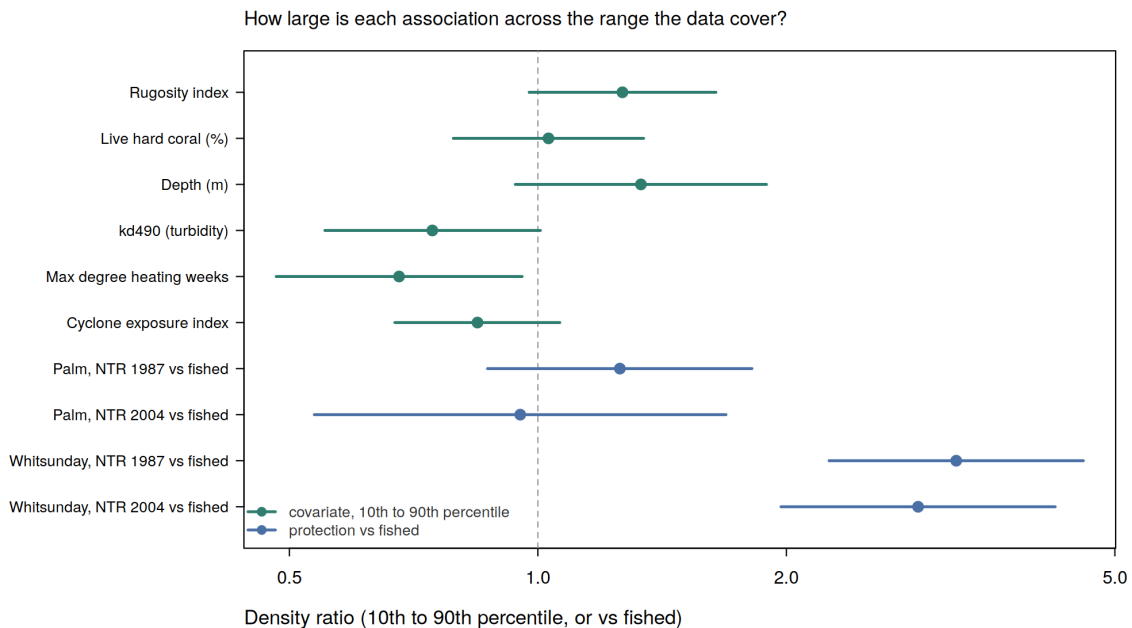


Figure 7. Effect magnitude across the observed range of each covariate, with the protection contrasts shown on the same scale.

Thermal stress gives the largest environmental magnitude at -32% across 0 to 3.4 degree heating weeks (0.68, 0.48–0.96) — deliberately the identified part of the curve, avoiding the boundary artefact. It is the only environmental term whose interval still excludes 1 and the only one to survive a multiplicity correction, which makes it the most suggestive of them rather than an established effect. Turbidity gives -25% (0.55–1.01) and cyclone exposure -16% (0.67–1.06); neither is distinguishable from no effect. Live coral and depth are not either.

The thermal result is suggestive, not established. Three things stand between it and a firmer reading. maxDHW carries 99% of its variance between region-years, so the term rests on roughly a dozen region-year contrasts rather than on 467 observations. Its worst-case concavity against the region-specific year trends is 0.951, so its coefficient is one possible split of variation those terms share rather than an independently identified effect. And those year smooths carry a basis-dimension warning that does not clear at $k = 6$, the largest the survey years allow, so one of the terms competing for that shared variation is itself unresolved. A negative association at moderate thermal stress is consistent with these data and worth pursuing with a design that can identify it. It is not demonstrated here, and it should not be quoted as a quantity without these three qualifications alongside it.

A note on where the two corrections disagree. Turbidity survives the Holm correction within its family (adjusted $p = 0.027$) but its magnitude interval includes 1. When two defensible corrections give different answers, the weaker reading is the one to quote, so turbidity is reported as unresolved rather than as an effect.

Two further cautions. The habitat magnitude does not survive the within-between split, as section 7 sets out. And the influence analysis in section 9 shows that turbidity and rugosity cross conventional significance depending on which single survey is omitted, so a precise interval on one contrast does not make either a stable finding. All of these statements are true at once, and they point the same way.

For scale: the Whitsunday protection contrast is $+221\%$. No environmental effect approaches it.

9. Sensitivity

Four checks specified before any were run, and one that was not. The plan names leave-one-region-year-out influence, restriction to the sheltered stratum, each region fitted alone, a comparison with and without the Whitsunday 2017 survey, and a distribution family swap. The first, second, third and fifth appear below; the fourth is covered by the leave-one-out analysis, which refits without Whitsunday 2017 along with the other twelve region-years.

The lagged cyclone term is exploratory, not prespecified. The plan asked for cyclone exposure integrated across survey intervals. This extract cannot support that: exposure is recorded only at survey points and the intervening years are absent. A lag over unequal survey gaps was substituted after the fact, and it is a weak substitute for integration over them. It is shown alongside the prespecified checks and labelled, rather than presented as one of them.

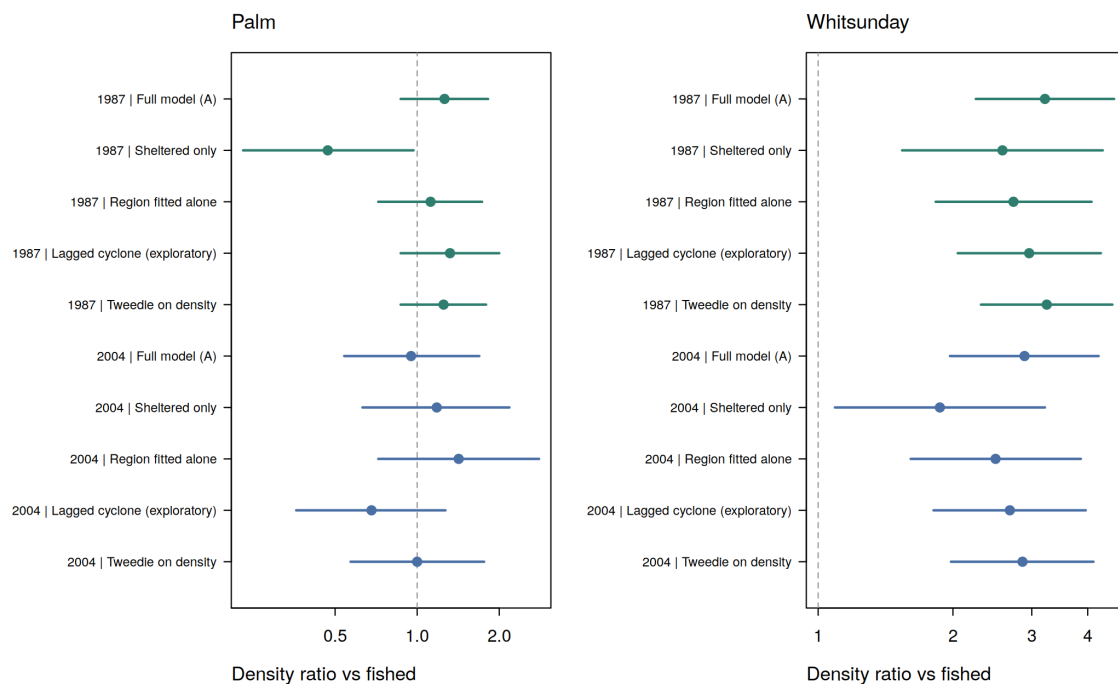


Figure 8. The protection estimate under five specifications.

	Full data	13 leave-one-out refits	5 specifications	Excludes 1?
Whitsunday NTR 1987	3.21	2.88 – 3.38	2.58 – 3.24	always
Whitsunday NTR 2004	2.89	2.59 – 3.09	1.87 – 2.89	always
Palm NTR 1987	1.26	1.16 – 1.31	0.47 – 1.31	in one of five
Palm NTR 2004	0.95	—	0.68 – 1.42	never

Dropping the Whitsunday 2017 survey raises the protection estimate to 3.35 rather than lowering it, which settles the concern raised by that survey's influence on the environmental terms.

Two things went the other way. The turbidity and rugosity terms both cross conventional significance depending on which single survey is omitted — kd_{490} ranges $p = 0.005$ to 0.126 , rugosity $p = 0.021$ to 0.192 . Neither is reliably distinguishable from zero and neither should be reported as an effect.

One result changes a conclusion. The Palm sheltered-only estimate for NTR 1987 is 0.47 (0.23–0.97) — below 1, though its interval reaches close to 1, where every other Palm specification gives 1.1 to 1.4. It rests on six fished and five reserve sites. Section 10 of the project plan set the rule for this check before any of it was run: the restriction tests the same thing as the adjustment without assuming the functional form is right or that exposure is the only cross-sectional confounder, so “if it agrees with the adjusted estimate, say so; if it does not, distrust the adjusted one.” It does not agree. Applying the plan's own rule, the Palm result is unresolved rather than null, and this report no longer describes it as a null.

What that does not do is explain the reversal. Eleven sites cannot distinguish a real local difference from noise, and no mechanism is proposed here. The honest statement is that Palm's two specifications disagree and the design cannot say which is right — which is itself a finding about what this dataset can support.

Two limitations of the restriction should be stated. The sheltered stratum contains only three fished sites in Whitsunday, so the restricted interval (1.54–4.32) is wider than the adjusted one; it corroborates the main estimate rather than replacing it. And the cyclone integration originally planned is impossible here — exposure is recorded only at survey points and the intervening years are absent from the extract. A lagged term was used instead and was not significant ($p = 0.149$).

10. Spatial structure in the residuals

If the site random effect has absorbed the spatial signal, site-level residuals should show no spatial pattern. This was tested within each region rather than across both, because a global test would only detect the 200 km gap between them. Inverse-distance weights, 4999 permutations.

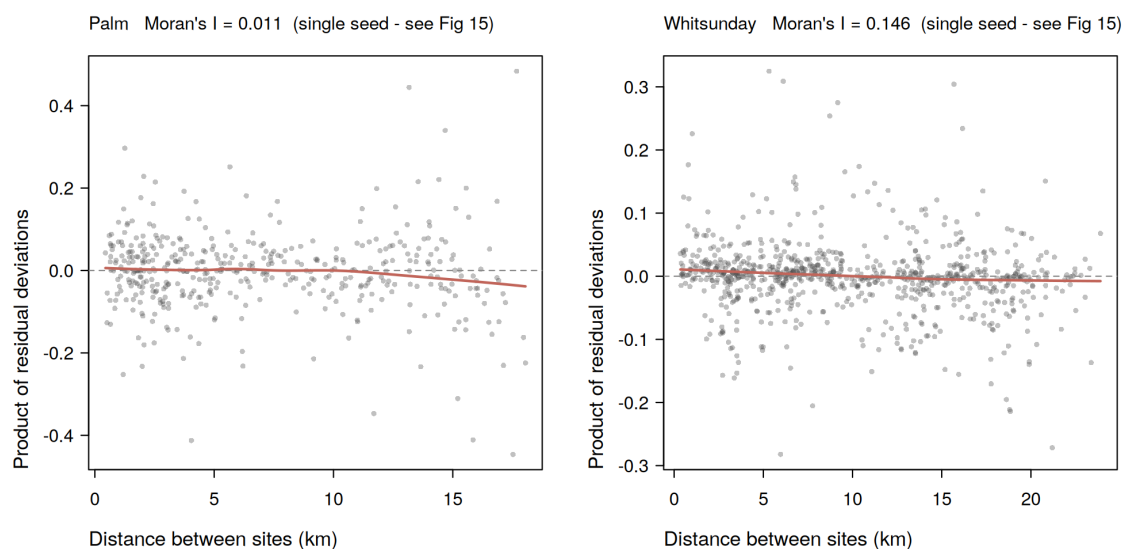


Figure 9. Product of residual deviations against distance between sites, by region, with Moran's I from a permutation test.

Randomised quantile residuals draw from a uniform, so anything computed from them depends on the random seed. Every earlier script fixed the same seed and none checked a second one. Across twenty seeds, Whitsunday's Moran's I is positive every time (0.079 to 0.167, median 0.114) but the permutation p-value falls below 0.05 in only eleven of twenty. Palm ranges -0.055 to 0.026 and is never significant.

A correction. An earlier version of this report quoted $I = 0.146$, $p = 0.007$ for Whitsunday from a single seed, and concluded that the site random effect had failed to absorb spatial structure. That

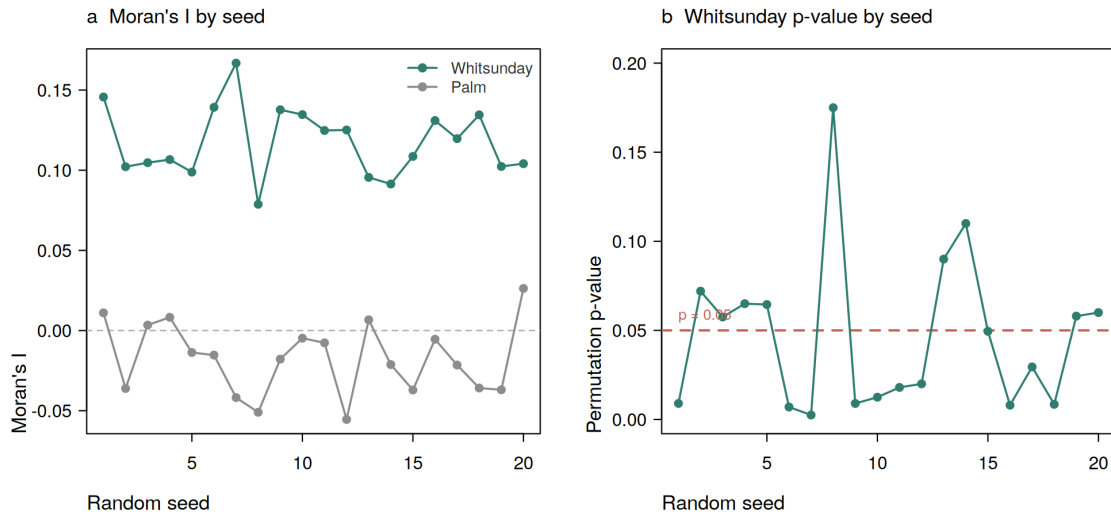


Figure 10. Moran's I and its permutation p-value across twenty random seeds. The dashed line is $p = 0.05$.

seed sits near the top of the range and the conclusion was too strong. What the evidence supports is a consistent difference in sign and magnitude between the two regions, not a significant result in one of them.

The reading that survives: residual spatial structure in Whitsunday is plausible and worth investigating with a spatial term, and there is no indication of any in Palm. It is not established at conventional thresholds and should not be reported as though it were.

This is documented rather than acted on either way. A spatial field was ruled out at the planning stage on the argument that a continuous field across two clusters 200 km apart would largely reproduce a region effect. That argument holds between regions and does not hold within Whitsunday, so the reasoning was too broad. Adding one now would be a substantial change made at the end of the project; it belongs at the top of any second version.

11. Goodness of fit: two questions, one bootstrap

Two things had never been checked: whether the negative binomial accounts for the observed zeros, and whether a site random intercept is enough to account for dependence between repeat visits to a site. Both ask whether data generated by this model would look like the data in hand, so both are answered from one parametric bootstrap. Each replicate draws a fresh set of site effects, simulates a dataset, refits the same model to it, and recomputes the statistic from the refit exactly as it is computed from the real fit.

Three things in that sentence are corrections to an earlier version. It refits, rather than simulating at fixed means, so the estimation of the parameters is inside the reference distribution. It redraws the site effects from their estimated distribution rather than reusing the fitted values, which had understated between-site variation — the fitted site effects have a standard deviation of 0.219 against an estimated component of 0.307 — and had made the null too narrow. And the residual-correlation statistic is the most negative correlation across all eleven survey gaps, so the search across gaps is part of the statistic rather than a choice made after seeing the answers; it is compared against its own simulated null, which already contains the negative correlation a site random intercept induces by construction.

Zeros: a real excess. The data hold about 16 more zeros than the fitted model expects, against a simulated distribution centred near 1 ($p = 0.003$). The negative binomial is not a complete description of the zero process here.

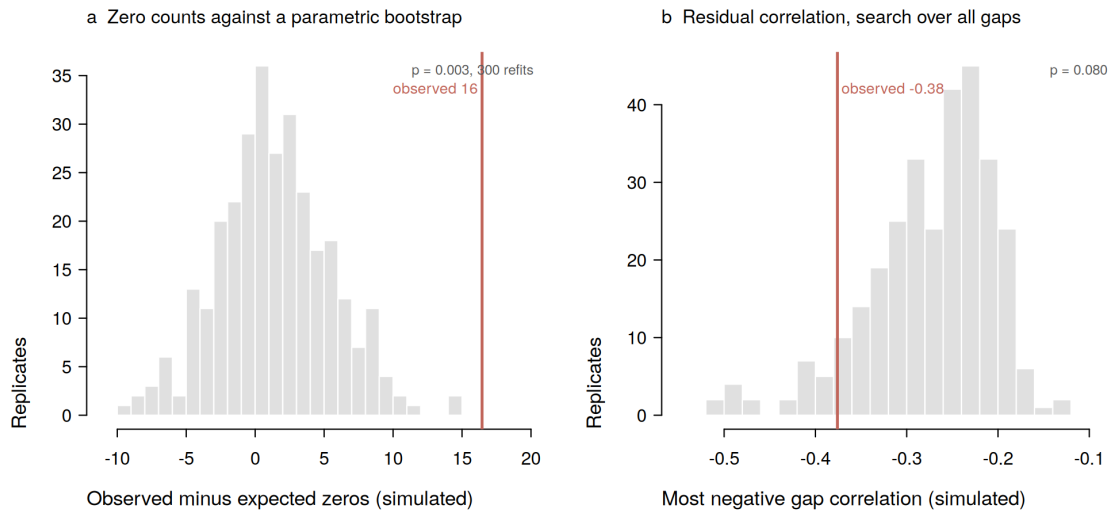


Figure 11. Each statistic against its own bootstrap null, from 300 simulate-refit replicates. Left: observed minus model-expected zeros. Right: the most negative residual correlation across all survey gaps.

What follows from that is less than it may seem. The two sensitivities available — trimming the mostly-zero site, and giving sites their own time trajectory — did not materially change the estimate, and there is a general argument that a zero excess inflates apparent overdispersion and so widens intervals rather than narrowing them. But no hurdle or zero-inflated negative binomial was fitted, and that refit is what would settle the question. The claim made here is the narrow one: the checks that were run did not move the estimate. Not that the distributional misfit is harmless.

Residual temporal correlation: unresolved, and exploratory. The most negative gap correlation is -0.376 , against a simulated null centred on -0.259 with a 95% interval of -0.452 to -0.176 . The observed value sits toward the extreme of that null without being clearly outside it, so the check neither confirms nor rules out residual temporal structure. It is left unresolved rather than pushed to one side. The exact simulated p is in the tables; it is deliberately not carried into this narrative, because reading a number from an exploratory diagnostic as a decision would be to treat it as the formal test it is not. An earlier version reported this as a clear finding, from testing the raw correlation against zero, selecting the strongest of eleven gaps after seeing them, and simulating without redrawing the site effects. All three made the test more permissive.

No mechanism is proposed for what remains. This analysis has no covariate that would distinguish a disturbance falling inside a survey gap from any other source of the same pattern, and the survey design cannot separate them. The finding is a statement about model adequacy — whether a site random intercept suffices — and nothing beyond that. More replicates would sharpen the p -value but would not change any conclusion drawn here, so none were added.

Neither diagnostic moves the protection estimate. Trimming the one site that is mostly zeros gives 3.27 (2.31 – 4.63) against 3.21 (2.25 – 4.58); adding a site-level random slope on year, so that sites have their own trajectory rather than only their own level, gives 3.27 (2.10 – 5.08). The wider interval on the last is the honest cost of the extra flexibility, not a change in the estimate.

12. What this does and does not establish

Defensible

- That *P. leopardus* density is roughly three times higher inside Whitsunday no-take zones than on fished sites in the same region, and that this is robust to exposure adjustment, exposure restriction,

region-separate fitting, distributional choice, the removal of any single survey, a multiplicity correction within its hypothesis family, and the trimming of the mostly-zero site.

- That the protection association differs between the two regions — the Q3 result, and the one that clears its multiplicity correction by an order of magnitude.
- That the site random effect has the largest point estimate of any block dropped in turn, at 9.76 percentage points of deviance explained. Its bootstrap interval overlaps those of environment and the regional year trends, so the ordering among blocks is not established. H4 is supported in direction; it is not demonstrated as a ranking.
- That no clear association was found for the three habitat variables chosen before modelling — rugosity, live hard coral and depth — once site and time are accounted for. This is a null for those variables as measured in this extract, not a finding that habitat does not predict density.
- That the data are consistent with a negative association between moderate thermal stress and density, of about -32% across the identified range. This is suggestive rather than established: the term rests on roughly a dozen region-year contrasts, its worst-case concavity against the region-specific year trends is 0.951 so it is one possible split of shared variation, the year smooths it competes with carry an unresolved basis warning, and the high end of the curve is not identified. It belongs in the list of things worth pursuing with a design that can identify them, not in the list of things shown.
- That Moran's I on residuals is consistently positive in Whitsunday and consistently near zero in Palm, across every randomisation tested — though not that the Whitsunday value is significant, which depends on the seed.

Not supportable

- That the Palm region has no protection effect. Its specifications disagree, and the one the plan nominated as most trustworthy points the other way. Unresolved is the correct word, and it is not the same as null.
- Any quantitative statement about how much of the variation management explains. Protection changes within none of the 71 sites, so it and the site random intercept draw on the same between-site degrees of freedom, and this design yields no unique, assumption-invariant share. The out-of-sample result in section 6 is a predictive statement and cannot be converted into one.
- That the Whitsunday difference is an effect of protection. It is a robust adjusted association. Measured wave exposure did not account for it under either adjustment or restriction, which is not the same as ruling exposure out — the categorisation is coarse and the sheltered stratum broad. Whatever else distinguished the sites that were zoned is not addressed at all, and cannot be with these data.
- Any mechanism for the residual temporal correlation, which is in any case unresolved.
- Any causal statement. Protection was not assigned at random. Adjusting for exposure addresses one measured confounder, imperfectly and by a coarse categorisation, and leaves every unmeasured one untouched.
- Any explanation for why the two regions differ. The difference is documented; the cause is not in these variables.
- Any claim about fishing pressure or effort.
- Anything about biomass, size structure, recruitment or stock status.
- Extrapolation to offshore reefs, to other regions, or to *P. maculatus*.

- An effect for any of the three prespecified habitat variables. Neither component of the within-between split excludes 1, so the pooled magnitude should not be quoted on its own. This says nothing about habitat measures the extract does not contain.

Conditional on choices made in advance

The habitat null holds for the specific covariates selected before modelling — rugosity, live hard coral and depth. LCC_% was considered as an alternative coral measure and deliberately not substituted after the null appeared, because changing a covariate in response to a null result is selection on the outcome. The null should be read as conditional, not as a general statement that habitat does not matter.

13. Questions this analysis could not answer

- Is transect-level data with species-specific lengths available, and through what route? Species-level biomass and size structure are the binding limitation on anything fisheries-relevant here.
- What survey area underlies the density values, and is the change in scaling convention between 2017 and 2018 intentional?
- Is external cyclone track data available, so exposure can be integrated across survey intervals rather than read at survey points? Exposure is currently read at survey points only, so any disturbance falling between two surveys is invisible to the covariate. This is a stated limitation of the covariate rather than an inference from a result.
- Is the residual spatial structure in Whitsunday recognised in AIMS' own analyses, and is there a site-level covariate — reef geomorphology, aspect, proximity to channels — that would account for it?
- For Keppel 2021, legal density exceeds total density at 17 of the 18 sites surveyed, by factors of 2.1 to 6.8, and the implied mean weight of legal-sized fish is 0.30 kg against 1.10–2.17 kg in every other region-year. Total density for those rows recovers cleanly on the same scaling step, so the survey area is unchanged, and the discrepancy is not a constant multiple. (For completeness: the whole extract holds 18 such rows. The eighteenth is Whitsunday 2018 site HY3, where the excess is 0.0018 — a rounding artefact, not the same phenomenon.) Is this a known processing issue?

Scope and standing

An observational analysis of a public monitoring dataset, carried out as a self-directed exercise. Nothing here is causal: protection was not assigned at random and the data cannot support a causal claim.

Every figure and number is reproducible from six R scripts using base R and mgcv, with nothing to install. The numerical outputs were verified across two independent clean runs from an empty output directory, which produced byte-identical tables, logs and figures. Wording in the scripts has been revised since that verification without rerunning, so the current text of the logs has not itself been re-verified byte for byte; every number in them is the output of those two runs. A decisions log records every change made to the analysis plan, what prompted it, and the cases where something was deliberately not changed.

This is the fourth revision. Successive audits against the project's own planning documents withdrew two attempts to quantify how much management explains, widened every interval, rebuilt two goodness-of-fit tests that had been too permissive, corrected the out-of-sample comparison in section 6 for clustering and for the treatment of unseen-site random effects, corrected two factual errors and a contradiction between tables naming the same model, and ran diagnostics that had

been missed. Where a conclusion changed, the earlier version is stated alongside it rather than quietly replaced — including one case where a correction moved a result from clearly significant to unresolved.

Reproducibility

Six scripts, run in order, regenerate every figure and number from the raw CSVs: coral_trout_stage1.R, then stage2_step1 through step5. The numerical outputs have been verified across two independent clean runs from an empty output directory, which produced byte-identical tables, logs and figures; Step 2 was verified the same way again after its cross-validation was corrected. Step 2 is the slow one — 200 site-level bootstrap resamples plus ten cross-validation folds — and Step 5 simulates and refits 300 times; all resample indices and simulation seeds are drawn up front under fixed seeds, so results do not depend on how many cores do the work.

One qualification on that. Descriptive wording inside the scripts has been revised since those runs, without rerunning them, and the corresponding lines of the committed logs were edited to match. So the current text of the logs has not itself been re-verified byte for byte against a fresh run. The edits were prose only — a diff of the logs before and after contains no numeric token — and every number in them remains the output of the verified runs. Rerunning would confirm the text; it was not done because it would change nothing but the text.

Nothing needs installing: the analysis uses base R and mgcv, which ships with every R installation. That is not the same as having no versions to reconcile. mgcv's version tracks the R version, and REML fitting and the nb() family have both changed across releases, so every log records the R version, the mgcv version and the platform that produced it. DECISIONS.md records every change made to the analysis plan, what prompted it, what was deliberately not changed, and — in one case — an audit finding that turned out to be wrong.

Data source and attribution

The source data for this report are from the Australian Institute of Marine Science (AIMS) dataset cited below. This analysis uses the Palm and Whitsunday surveys from 2007 to 2018. The source files are not redistributed with this report; data access and reproduction instructions are provided in the accompanying repository's DATA.md.

Australian Institute of Marine Science (AIMS). (2022). *Spatio-temporal dynamics of coral reef fish assemblages on inshore reefs of the Great Barrier Reef*. <https://apps.aims.gov.au/metadata/view/814a0be3-ed85-4a43-87b7-59ece6eb6a05> (metadata record accessed 2 September 2026).

The dataset is licensed under Creative Commons Attribution 3.0 Australia (CC BY 3.0 AU). The data were subsetted and transformed for the statistical analyses, figures and tables presented in this report. Based on Australian Institute of Marine Science data.